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# AIM:- Face Detection
# PROGRAM:-
import cv2
import matplotlib.pyplot as plt
import os # Import os module to check for file existence

# Load the pre-trained Haar Cascade classifier for frontal face detection
# Ensure this XML file is available. It's usually part of OpenCV installations.
face_cascade = cv2.CascadeClassifier(cv2.data.harcascades + 'haarcascade_frontalface_default.xml')

# Path to the image file in Colab environment
image_path = "PICTUTE.png"

# Check if the image file exists. If not, use a known available image as a fallback.
if not os.path.exists(image_path):
    ... print(f"Warning: Image '{image_path}' not found. Please upload it to your Colab environment if yo
    ... print("Falling back to 'V KOHLI.jpeg' for demonstration.")
    ... image_path = "V KOHLI.jpeg" # Fallback to the known available image

try:
    ... # Read the image
    ... img = cv2.imread(image_path)

    ... # Check if image is loaded successfully (even after potential fallback)
    ... if img is None:
    ...     ... raise FileNotFoundError(f"Error: Could not load image from {image_path}. It might be corrupte

    ... # Convert the image to grayscale, as face detection works best on grayscale images
    ... gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    ... # Detect faces in the grayscale image
    ... # scaleFactor: Specifies how much the image size is reduced at each image scale.
    ... # minNeighbors: Specifies how many neighbors each candidate rectangle should have to retain it.
    ... # minSize: Minimum possible object size. Objects smaller than that are ignored.
    ... faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30, 30))
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...# Draw rectangles around the detected faces
...for (x, y, w, h) in faces:
...    cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2) # Green rectangle with thickness 2

...# Display the result using matplotlib
...plt.figure(figsize=(10, 8))
...plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)) # Convert BGR to RGB for correct display
...plt.title(f'Face Detection on {os.path.basename(image_path)}')
...plt.axis('off') # Hide axes
...plt.show()

...if len(faces) > 0:
...    print(f"Detected {len(faces)} face(s) in the image '{os.path.basename(image_path)}'.")
...else:
...    print(f"No faces detected in the image '{os.path.basename(image_path)}'.")

except FileNotFoundError as e:
...    print(e)
except Exception as e:
...    print(f"An unexpected error occurred: {e}")

# Note: cv2.VideoCapture(0), cv2.imshow, cv2.waitKey, and cv2.destroyAllWindows
# are typically used for real-time video processing and display on a local machine
# and do not function in Google Colab environment directly. For displaying results,
# matplotlib.pyplot is used.
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Warning: Image 'PICTUTE.png' not found. Please upload it to your Colab environment if you wish to use it.
Falling back to 'V KOHLI.jpeg' for demonstration.

Face Detection on V KOHLI.jpeg



Detected 2 face(s) in the image 'V KOHLI.jpeg'.